0.1 Getting Started with Raspberry Pi

Edge AI Engineering Bootcamp | Analog Data

From Unboxing to SSH — A Complete Setup Guide

What You'll Need Before Starting

Item	Notes	
Raspberry Pi board	Pi 4B, Pi 5, or Pi Zero 2W	
microSD card	16GB minimum, 32GB+ recommended, Class 10 or better	
microSD card reader	Built into your laptop, or a USB adapter	
Power supply	USB-C for Pi 4B/5 (needs 5V/3A), micro-USB for older boards	
Computer	Windows, Mac, or Linux — used only for setup	
Internet connection	For downloading the OS and updates	
(Optional) Monitor, keyboard, mouse, HDMI cable	Only needed if you're **not** going headless	
(Optional) Ethernet cable	Useful for the most reliable first connection	

Key idea: Once setup is done, you often won't need a monitor or keyboard connected to the Pi at all — you'll control it entirely from your laptop over the network. This is called running the Pi **"headless."**

Step 1: Download and Install Raspberry Pi Imager

Raspberry Pi Imager is the official tool for writing (also called "flashing") an operating system onto your microSD card.

1. Go to **raspberrypi.com/software**
2. Download **Raspberry Pi Imager** for your computer's OS (Windows/Mac/Linux)
3. Install and open it

```
┌──────────────────────────────────────┐
│          Raspberry Pi Imager         │
│                                      │
│  [ CHOOSE DEVICE  ]                  │
│  [ CHOOSE OS      ]                  │
│  [ CHOOSE STORAGE ]                  │
│                                      │
```

```
│               [ NEXT ]                │
└───────────────────────────────────────┘
```

Step 2: Choose Your Device, OS, and Storage

1. **Choose Device** → select your exact board (e.g., "Raspberry Pi 4")
2. **Choose OS** → select an operating system:

OS Option	Best For
Raspberry Pi OS (64-bit)	General use, most beginner projects — recommended default
Raspberry Pi OS Lite (64-bit)	No desktop GUI — perfect for headless/server-style setups, saves resources
Raspberry Pi OS (Legacy)	Older software compatibility
Other OS (Ubuntu, etc.)	Specific project needs

For this course: use **Raspberry Pi OS Lite (64-bit)** if you're comfortable with the command line, since we'll be working headless and don't need a desktop environment eating up RAM.

3. **Choose Storage** → select your microSD card (double-check you've picked the right drive — this step erases everything on it)

Step 3: Configure Before Flashing (Very Important)

Before you click "Write," click the ⚙️ **gear icon** (or you'll be prompted automatically) to open **Advanced Options**. This step is what lets you set up SSH *before* the Pi ever boots — no monitor or keyboard needed.

Configure the following:

```
┌──────────────────────────────────────────┐
│              Advanced Options            │
│                                          │
│  ☑ Set hostname:      mypi.local         │
│  ☑ Enable SSH                            │
│      ● Use password authentication       │
│      ○ Allow public-key authentication   │
│        Username: pi                      │
│        Password: ********                │
│                                          │
│  ☑ Configure wireless LAN                │
│    SSID:     YourWiFiName                │
│    Password: YourWiFiPassword            │
│    Country:  IN                          │
│                                          │
│  ☑ Set locale settings                   │
│    Timezone: Asia/Kolkata                │
```

```
│      Keyboard layout: us                │
└─────────────────────────────────────────┘
```

Why this matters:

- **Enable SSH** — without this, SSH is disabled by default on a fresh install, and you won't be able to connect remotely at all.
- **Set username/password** — newer Raspberry Pi OS versions no longer use a default `pi` / `raspberry` login for security reasons — you must set your own.
- **Configure WiFi** — lets the Pi connect to your network automatically on first boot, so you don't need Ethernet or a monitor.
- **Set hostname** — lets you find the Pi on your network by name (e.g., `mypi.local`) instead of hunting for its IP address.

Once configured, click **Save**, then **Write**, and confirm. This will take a few minutes.

Step 4: First Boot

1. Once flashing finishes, safely eject the microSD card
2. Insert it into the Raspberry Pi
3. Connect power (and Ethernet, if using it)
4. Wait 60–90 seconds for the first boot — the Pi is expanding the filesystem and applying your settings

LED check (most boards):

- Solid red = power is connected
- Blinking green = the SD card is being read (normal activity)

Step 5: Find Your Pi's IP Address

You need to know your Pi's address on the network to connect to it. A few ways to find it:

Option A — Use the hostname (easiest):

```
ping mypi.local
```

If it responds, you can connect directly using `mypi.local` instead of an IP address.

Option B — Check your router's admin page Log into your WiFi router (usually `192.168.1.1` or `192.168.0.1`) and look for connected devices — the Pi should appear by hostname.

Option C — Use a network scanner Tools like `nmap` (Linux/Mac) or **Angry IP Scanner** (Windows) can list all devices on your network.

```
nmap -sn 192.168.1.0/24
```

Step 6: Connect via SSH

SSH (Secure Shell) lets you control the Pi's command line remotely from your own computer, over the network, as if you were typing directly on it.

```
  Your Laptop                        Raspberry Pi
┌─────────────────┐               ┌─────────────────┐
│   Terminal      │  ── SSH ──▶   │  Command Line   │
│   (you type)    │   (network)   │  (Pi executes)  │
└─────────────────┘               └─────────────────┘
```

On Windows

Use **PowerShell**, **Command Prompt**, or **Windows Terminal** (SSH is built in on modern Windows):

```
ssh yourusername@mypi.local
```

On Mac/Linux

Open **Terminal**:

```
ssh yourusername@mypi.local
```

If hostname doesn't resolve, use the IP address instead:

```
ssh yourusername@192.168.1.45
```

First-time connection: You'll see a security warning like this — this is normal, type `yes`:

```
The authenticity of host 'mypi.local' can't be established.
Are you sure you want to continue connecting (yes/no)?
```

Then enter the password you set in Raspberry Pi Imager. If it's correct, you'll see a prompt like:

```
yourusername@mypi:~ $
```

Congratulations — you're now controlling the Raspberry Pi remotely, entirely through the command line.

Step 7: First Things to Do After Connecting

1. Update the system:

```
sudo apt update && sudo apt upgrade -y
```

This refreshes the list of available software and updates everything already installed.

2. Check basic system info:

```
hostname -I        # shows the Pi's IP address(es)
uname -a            # shows OS/kernel version
vcgencmd measure_temp   # shows CPU temperature
df -h               # shows storage space used/free
free -h             # shows RAM usage
```

3. Change the hostname or password later (if needed):

```
sudo raspi-config
```

This opens a menu-based configuration tool where you can change almost any setting — hostname, password, locale, interfaces (camera, I2C, SPI), and more.

Common Problems & Fixes

Problem	Likely Cause	Fix
`ssh: connect to host mypi.local port 22: Connection refused`	SSH wasn't enabled during flashing	Re-flash with SSH enabled, or add an empty file named `ssh` (no extension) to the boot partition manually
Can't find `mypi.local`	mDNS/hostname resolution issue (common on some Windows setups)	Use the Pi's IP address directly instead
WiFi not connecting	Wrong SSID/password, or 5GHz-only network on an older Pi that only supports 2.4GHz	Double-check credentials; connect to a 2.4GHz network if using an older board
Permission denied (password)	Wrong username/password	Re-check what you set in Imager; usernames are case-sensitive
Pi not appearing on network at all	Power issue or bad SD card write	Try a different power supply/cable; re-flash the SD card

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Quick Recap

1. **Flash the OS** using Raspberry Pi Imager — choose device, OS, and storage
2. **Configure SSH, WiFi, username/password, and hostname** in Advanced Options *before* writing — this enables headless setup
3. **Boot the Pi** and wait for it to connect to your network
4. **Find its address** using the hostname (`mypi.local`) or by scanning your network

5. **Connect via SSH** from your computer's terminal
6. **Update the system** and explore basic commands to confirm everything works

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Self-Check Questions

1. What does "headless setup" mean, and why is it useful?
2. Why do you need to enable SSH *before* flashing, instead of after the Pi has booted?
3. What's the difference between connecting using a hostname (`mypi.local`) versus an IP address?
4. What command would you use to check how much RAM and storage your Pi currently has free?
5. If `ssh yourusername@mypi.local` gives "Connection refused," what are two possible causes, and how would you fix each?